Adrián Berenguer Agustí

Valencia • Spain • adrian9420agusti@gmail.com • LinkedIn • Github

Bioinformatician working on long-read transcriptomics and genomics within the SQANTI ecosystem. Research interest in isoform-resolved transcriptomics and DNA methylation applied to Alzheimer's disease.

Education

University of Valencia — Valencia, Spain
Master's Degree in Bioinformatics. 2nd highest mark. 90 ECTS. GPA 9.09 out of 10. — 2025–2027
Thesis: Reference-independent SQANTI tool for non-model organisms.
Relevant Coursework: Honors (3) in Computers and operating systems, Structural Bioinformatics and Medicine and clinical trials.

Bachelor's Degree in Biotechnology. GPA 8.05 out of 10. — 2021–2025
Thesis: Reduction of ethanol content in wine yeast using ALE and CRISPR-Cas9 systems.
Relevant Coursework: Honors (11) in Bioinformatics, Economy, Bioreactors, Immunology, Legal aspects, Metabolism, Molecular biology, Molecular genetics, Molecular analysis, Industrial biotechnological products and Molecular plant biology.

Benicalap High School — Valencia, Spain
GPA 10 out of 10. Honors. — 2019–2021

Experience

Institute of Integrative Systems Biology UV-CSIC — Valencia, Spain
Internship & Master's thesis at Genomics of Gene Expression Lab. PI: Ana Conesa. — 06/26 – Present

- Development of a multi-sampling filter for the SQANTI long-reads tool, merged into the official release. Python & R.
- Development of SQANTI free-reference annotation tool. Python & R.

Institute of Agrochemistry and Food Technology CSIC — Valencia, Spain
Studentship JAE INTRO — 09/25 – 02/2026

- Phenotyping of aging and neuromuscular function in a *C. elegans* model fed on postbiotics: longevity, motility, and fertility assays. Statistical analysis of survival curves.

Institute of Integrative Systems Biology UV-CSIC — Valencia, Spain
Bachelor's thesis — 09/24 – 06/2025

- CRISPR-Cas9 system for making point mutations in wine yeasts that are low ethanol producers.
- Development of a new wine yeast strain by directed evolution that produces more glycerol and less acetic acid.
- Protein docking for evaluating point mutations that affect protein-protein interactions. Python and PyMol.

Computational Skills

Programming: Python advanced level, Bash advanced level, R medium level, SQL basic level.

Environments: Linux, shell scripting, Git and GitHub.

Transcriptomics: Long-read RNA-seq quality control, isoform-level annotation and filtering (SQANTI).

Python libraries: NumPy, pandas, SciPy, Matplotlib, Biopython.

R libraries: tidyverse, ggplot2, Bioconductor, DESeq2, survival.

Machine learning: scikit-learn, PyTorch, TensorFlow.

High-performance computing: SLURM job scheduling on HPC clusters.

Additional Training

Machine Learning and Big Data for Bioinformatics. University of Granada.

Building AI - Advanced. University of Helsinki.

Scientific production

Software:

- SQANTI3. Contributor to release v6.1. Multi-sample filtering module for long-read isoform quality control. ConesaLab/SQANTI3

Submitted papers:

- Berenguer-Agustí, A., Hernández, P., Ortega, L., Parodi, A., Benítez-Páez, A. (2026). *Caenorhabditis elegans* as a model for testing microbiota-derived nutritional solutions for human health. Nutrition Reviews (Oxford Academic). Under review.
- Garrigós V., Muñoz-Margaix H., Kojenova G., Miró-Tudela M., Berenguer-Agustí A., Picazo C., Aranda A., (2026). Systematic mutational analysis reveals functional redundancy and pathway crosstalk in succinic acid biosynthesis by *Saccharomyces cerevisiae* during winemaking. Journal of Agricultural and Food Chemistry. Under review.

Papers in preparation:

- Alberto, M., Berenguer-Agustí, A., Vázquez R., Meca, G., Ferrer, S., Pardo, I., Arena, M. (2026). Spray-dried supernatant preparation from wine-associated lactic acid bacteria inhibit *Brettanomyces bruxellensis* growth, adhesion, and biofilm formation and show antioxidant capacity. Manuscript in preparation.

Posters:

- Hernández, R., Berenguer-Agusti, A., Alonso, L., Dasí, F. (2026). The metabolic panacea? Evaluation of the effectiveness and safety of semaglutide indicated for weight loss in obese patients. XIV biomedical research congress, Valencia, Spain.

Oral presentations:

- Berenguer-Agusti, A., Benítez-Páez, A. (2026). *C. elegans* model for strains with postbiotic potential. Workshop JAE Intro Conexión Microbioma-CSIC.

Awards

2026

1st place National Open Innovation competition.

1st place National BISC-E Challenge.

Finalist at European BISC-E Challenge.

2nd place Valencia Santander X awards.

2nd-3rd best National Bachelor Thesis in Sustainability.

Selected for Generación Propósito programme. Fundación Princesa de Girona

2025

Finalist in Valencian Young Biotech Talent.

1st place Agrotechnology Competition University of Valencia.

1st place in the Sciences Motivem Awards.

1st place pre-incubators University of Valencia.

1st place economy pre-incubator University of Valencia.

2024

1st place Agrotechnology Competition University of Valencia.

2nd place University of Valencia Startup Award.

2021

6th place Biology Olympiad.

2nd place Valencian Community Sapiencia Awards.

Science communication

- Speed Dating (2026). European Researchers' Night.
- Microbiome's talk (2026). Pint of Science.
- Volunteer at International Society for Computational Biology Regional Student Group in Spain.

Languages

- English advanced C1 – Spanish Official School of Languages.
- French basic A2 – Spanish Official School of Languages.
- Catalan advanced C1 – Valencian Language Certification Board.
- Spanish – Native.